

QUANTIFICATION OF FAMILIAL PREDISPOSITION TO DISEASE

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Summary: The article sets out to develop an index for quantifying the susceptibility or resistance to disease conditions, as manifested in given families, i.e. an index of familial predisposition. For this purpose all relevant family parameters are employed and weighted according to their relative importance in the manifestation of the disease. Apart from being based on actual disease manifestation, the family risk index also takes into account the potential of developing the disease inherent in non-affected family members (either alive or dead due to other causes). Based on the devised index a bootstrap test for the presence of susceptibility and/or resistance in a sample of families is proposed. The plausibility of the statistical procedure is illustrated on a family data set, scored for coronary heart disease.

1. INTRODUCTION

The concept of a positive (or negative) "family history" has found widespread application in genetic epidemiology. In general, it denotes an unusually high (or low) aggregation of individuals with a certain disease con-

dition or abnormality, within a particular family (BMJ (1977a), BMJ (1977b), Saunders (1982), Berg (1983), Harnden (1983), Saunders & Williams (1983), Mulvihill (1985) and Williams (1988)). Several studies demonstrate such familial aggregation or clustering, and evaluate the role of family predisposition in a variety of conditions, for example in the case of coronary heart disease (BMJ (1977a), Berg (1983) and Williams (1988)), in the case of cancer (Petrakis et al. (1982), Schneider et al. (1983) and Sellers et al. (1987)), and in the case of alcoholism and alcohol-related problems (BMJ (1977b), Saunders (1982), Saunders & Williams (1983), Volicer et al. (1985) and Schuckit (1985)). Attempts have also been made to evaluate and determine *the degree* of aggregation or susceptibility (Williams (1988), Snowden et al. (1982), Schneider et al. (1983), Sellers et al. (1987) and Schuckit (1985)). The difficulties encountered in this approach relate to the fact that the concept of family history or familial predisposition requires a rigorous definition, as well as an objective weighting of its underlying variable components, such as family size, number of individuals affected, the sex and age structure of the family, the effect of these on disease manifestation, and the interrelationships of family members.

The central premise of the present study is the postulate of a familial susceptibility or predisposition to or, conversely, a familial non-predisposition or resistance to certain disease conditions. This implies that members of the susceptible families are at greater risk of being affected than members of the resistant families. On the other hand, families of "average" susceptibility are expected to be "randomly" affected.

Indeed, it is plausible to expect the degree of familial predisposition to some, especially multifactorial, diseases to be a continuous variable with a continuous probability distribution.

Against this background, and taking all the relevant familial parameters into account, a theoretical method has been devised to quantify the familial predisposition in families (Groeneveld & Hitzeroth (1986)). The objective of the present study is to present and elaborate this statistical procedure in greater detail with particular reference to

- the development and calculation of a realistic *risk index* which is specific to each family,
- testing with the aid of such risk indices, whether or not a familial predisposition to a disease is evident in a given population, and
- the classification of the families in a data set as either being susceptible, or resistant to a disease, or having an indeterminate familial predisposition.

2. THE DATA

For the purpose of this study a family is defined in terms of a specific index person (known to be either affected or non-affected) and his/her genetically related family members, as follows:

- 1st degree relatives (children, brothers/sisters and parents), and
- 2nd degree relatives (grandchildren, uncles/aunts and grandparents).

The data available for addressing the question of familial predisposition typically consist of a collection of family pedigrees and histories containing the required information. In the present study a sample of 206 families is available from a study of coronary heart disease (CHD) (see table I: Hitzerth, *et al.* (1991)). The following information was obtained for each family member: sex, current age, or age at death (if applicable), whether or not he/she had coronary heart disease and at what age this occurred, and finally, the genetic interrelationship to the index case and other family members. It was required to develop a method for quantifying the degree (the presence/absence) of familial predisposition to CHD in these families.

In our data the disease condition under consideration (coronary heart disease) manifests itself as a discontinuous binary variable, with no degrees of disease manifestation. In view of this, the family data in the above-mentioned format is inherently incomplete. Whereas the question of whether or not a particular family member is affected (and the age when this

Table I. Actual frequency distribution according to sex and age categories, of all individuals of a sample of 206 families scored for coronary heart disease (De Villiers, L.S., unpublished data).

Age categories (years)	MALES			FEMALES		
	Not affected by disease	Affected by disease	Total	Not affected by disease	Affected by disease	Total
1. 0-4	17	0	17	14	0	14
2. 5-9	9	0	9	10	0	10
3. 10-14	16	0	16	13	0	13
4. 15-19	22	0	22	12	0	12
5. 20-24	58	1	59	28	0	28
6. 25-29	63	6	69	41	1	42
7. 30-34	78	21	99	86	2	88
8. 35-39	93	37	130	89	6	95
9. 40-44	125	71	196	109	12	121
10. 45-49	137	61	198	113	16	129
11. 50-54	183	91	274	152	15	167
12. 55-59	166	70	236	164	32	196
13. 60-64	223	96	319	248	30	278
14. 65-69	197	65	262	225	36	261
15. 70-74	293	65	358	337	25	362
16. 75-	460	53	513	549	29	578

occurred) can usually be answered with certainty, it is not possible to establish whether or not a healthy individual will in his future life develop the disease or at what age this might occur. Even in the case of a family member having died due to another cause it may well be that he could have developed the disease at a later age had he remained alive. Hence, merely scoring the actual manifestation of the disease and the age of onset ignores the potential disease predisposition inherent in non-affected members (deceased or alive) and can lead to an underestimate of the overall family predisposition as expressed by the calculated risk index. Our statistical method for quantifying familial predisposition to CHD in terms of a risk index is devised to incorporate sex, age and the potential familial risk still inherent in the events of non-manifestation of CHD (see below).

Before the proposed risk indices can be calculated, the data must be summarised as in table II. The age stated for non-affected living individuals is their age at the time of recording, and for non-affected deceased individuals, their age at death. In the case of affected individuals (alive or deceased) the age of disease-onset (or disease presentation) is recorded.

3. THE DEVELOPMENT OF A RISK INDEX

We define p_{ij} as the probability of randomly developing the disease in the i -th age category and the j -th sex. To each affected individual in the data set the following relative weight is assigned:

$$W_{ijk\ell} = (1 - p_{ij}) \log \frac{(1 - p_{ij})}{p_{ij}} \text{ for the } \ell\text{-th individual in the } k\text{-th family.}$$

To each non-affected individual in the data set the following relative weight is assigned:

$$W_{ijkv} = -p_{ij} \log \frac{(1 - p_{ij})}{p_{ij}} \text{ for the } v\text{-th individual in the } k\text{-th family.}$$

These two assigned weights follow from the following two premises:

Table II. Theoretical frequency distribution of all individuals from a sample of families, according to sex and age categories, but irrespective of family relationship

Age categories (years)	MALES			FEMALES		
	Not affected by disease	Affected by disease	Total	Not affected by disease	Affected by disease	Total
1. 0-4	f_{111}	f_{112}	$f_{11\cdot}$	f_{121}	f_{122}	$f_{12\cdot}$
2. 5-9	f_{211}	f_{212}	$f_{21\cdot}$	f_{221}	f_{222}	$f_{22\cdot}$
3. 10-14	f_{311}	f_{312}	$f_{31\cdot}$	f_{321}	f_{322}	$f_{32\cdot}$
.	.	.	.	.	.	.
.	.	.	.	.	.	.
i.	f_{i11}	f_{i12}	$f_{i1\cdot}$	f_{i21}	f_{i22}	$f_{i2\cdot}$
.	.	.	.	.	.	.
15. 70-74	.	.	.	.	.	.
16. 75-	.	.	.	.	.	.
TOTAL	$f_{\cdot 11}$	$f_{\cdot 12}$	$f_{\cdot 1\cdot}$	$f_{\cdot 21}$	$f_{\cdot 22}$	$f_{\cdot 2\cdot}$

1. If the probability of random disease in a particular age-sex category decreases, the conditional probability that the disease is familial, having observed a diseased person, increases.
2. If the probability of random disease in a particular age-sex category increases, the conditional probability that the family is resistant, having observed a healthy person, also increases.

We now define the probability that the disease in a person in age category i , of sex j and family k is familial, given that the person has the disease, as $p_{ijk}^{(F)}|D$ and the probability that the person is resistant to the disease, given that he/she is healthy, as $p_{ijk}^{(R)}|\bar{D}$. The two premises can now be restated as follows:

1. If p_{ij} decreases, $p_{ijk}^{(F)}|D$ increases.
2. If p_{ij} increases, $p_{ijk}^{(R)}|\bar{D}$ also increases.

Consider for a person with the disease the following two hypotheses:

H_1 : The disease in the person is familial with probability

$$p_{ijk}^{(F)}|D > p_{ij}$$

H_2 : The person is resistant or the disease is random with probability

$$1 - p_{ijk}^{(F)}|D$$

According to the Kullback-Leibler information criterion (Mathai & Rathie (1975)), the statistic for discriminating between H_1 and H_2 is

$$p_{ijk}^{(F)}|H \log \frac{p_{ijk}^{(F)}|D}{1 - p_{ijk}^{(F)}|D}$$

Now, if p_{ij} decreases, $p_{ijk}^{(F)}|D$ increases (premise 1). Hence there is a monotonic increasing relationship between

$$\frac{1 - p_{ij}}{p_{ij}} \text{ and } \frac{p_{ijk}^{(F)}|D}{1 - p_{ijk}^{(F)}|D}$$

and thus also between

$$(1 - p_{ij}) \log \frac{(1 - p_{ij})}{p_{ij}} \text{ and } p_{ijk}^{(F)} | D \log \frac{p_{ijk}^{(F)} | D}{(1 - p_{ijk}^{(F)} | D)}$$

Therefore, since $p_{ijk}^{(F)} | D$ is unknown, $(1 - p_{ij}) \log \frac{(1 - p_{ij})}{p_{ij}}$ can be used as information for discriminating between the two hypotheses, given the relative weight for diseased individuals defined earlier in this section. Estimates of p_{ij} are available from the data, namely f_{ijl}/f_{ij} (table I), and they are used for calculating the relative weights.

For a healthy person, we consider the following two hypotheses:

H_1 : The person is resistant with probability $p_{ijk}^{(R)} | \bar{D} < p_{ij}$

H_2 : The disease in the person is familial or random with probability $1 - p_{ijk}^{(R)} | \bar{D}$.

Since, if p_{ij} increases, $p_{ijk}^{(R)} | \bar{D}$ also increases (premise 4), it follows that there is a monotonic increasing relationship between

$$p_{ij} \log \frac{p_{ij}}{(1 - p_{ij})} = -p_{ij} \log \frac{(1 - p_{ij})}{p_{ij}} \text{ and } p_{ijk}^{(R)} | \bar{D} \log \frac{p_{ijk}^{(R)} | \bar{D}}{(1 - p_{ijk}^{(R)} | \bar{D})}$$

Hence we can use $-p_{ij} \log \frac{(1 - p_{ij})}{p_{ij}}$ as information to discriminate between the two hypotheses and therefore as the relative weight for healthy persons, as defined previously. In practice we again use estimates of p_{ij} for calculating the relative weights. Whenever $p_{ij} = 0$, the relative weight does not exist. In this case a null weight is assigned, since it seems reasonable that no information is contained in $p_{ij} = 0$. The expected weight for the l -th individual in the age and sex category ij , and in the family k , can now be defined as follows:

$$E(W_{ijk\ell}) = \pi_{ij} (1 - p_{ij}) \log \frac{(1 - p_{ij})}{p_{ij}} - (1 - \pi_{ij}) p_{ij} \log \frac{(1 - p_{ij})}{p_{ij}}$$

This expected value will have the following values:

$$(i) E(W_{ijk\ell}) = 0,$$

if contracting the disease is a random event, since then $\pi_{ij} = p_{ij}$

$$(ii) E(W_{ijk\ell}) > 0,$$

if the individual has a higher than 'random' probability of developing the disease, since then $\pi_{ij} > p_{ij}$

$$(iii) E(W_{ijk\ell}) < 0,$$

if the individual has a lower than 'random' probability of developing the disease, since then $\pi_{ij} < p_{ij}$

The index of familial predisposition to a given disease condition for the k-th family is now defined as

$$\hat{I}_k = 1/n_k \sum_{ijk\ell} \hat{W}_{ijk\ell}$$

where $\hat{W}$ is an estimate of W from table I, where the summation extends over all members, affected and non-affected, of the k-th family, and where n_k is the number of individuals in the k-th family. Hence $E(\hat{I}_k) = 0$ for families where development of the disease occurs at random; $E(\hat{I}_k) > 0$ for families with a predisposition and $E(\hat{I}_k) < 0$ for families with a resistance to the disease.

In order to answer the question as to whether or not a familial predisposition to a disease exists in the sample of families, a statistic $\hat{Q}$ is defined as follows:

$$(3.1) \quad \hat{Q} = \sum_k \hat{I}_k^2,$$

with $E(\hat{Q}) = B$, say, when randomly affected by the disease, and $E(\hat{Q}) > B$ when a predisposition or a resistance to the disease exists. Hence $\hat{Q}$ can be used as a test statistic to distinguish between the following two hypotheses:

H_0 : The sample is derived from a population of families in which family members are randomly affected, i.e. there is neither a familial predisposition nor a resistance to the disease.

H_1 : The sample is derived from a population of families in which some families are predisposed to, and/or others resistant to the disease.

The quantity $\hat{Q}$ gives χ^2 -information for discriminating between H_0 and H_1 . It is not necessary to centralise this statistic under H_0 in the usual χ^2 fashion, because the non-central distribution under H_0 will be simulated.

The sampling distribution of $\hat{Q}$ under H_0 now needs to be established.

Since $\hat{Q}$ is defined for a sample of specific age, sex and family size structure as well as number of families its distribution will also depend on these variables. It will be impractical therefore, to determine sampling distributions for each possible case. It is proposed therefore, to rather generate an empirical sampling distribution for $\hat{Q}$, and to use this for the Bootstrap significance testing of H_0 . This process will be conducted as follows:

1. Tabulate the family data of the set under consideration according to the format of table I.
2. For each category in table I, calculate the empirical probabilities of being affected by the disease.
3. Using the probabilities calculated in step 2, randomly designate each individual in the data set as either affected or unaffected. For this purpose the actual status of individuals in the data set is ignored.
4. With the random allocations completed, and with due recognition of the original family groupings in the data set, calculate $\hat{I}_k$ for each family.
5. Then calculate $\hat{Q}$ in (3.1) and call the value obtained $\hat{Q}_n$.

6. Repeat steps 3, 4 and 5 a number of times, say $N = 200$ times.
 7. Calculate the empirical $100(1-\alpha)\%$ percentile of the N values of $\hat{Q}_0$.
 8. Calculate $\hat{Q}$ for the data set and use the percentile of the empirical distribution of $\hat{Q}_0$ calculated in step 7 to test H_0 at the α level of significance.
- Since each family index $\hat{I}_k$ is also calculated N times under H_0 it is possible to determine an empirical sampling distribution of $\hat{I}_k$ for each family. Percentiles of these distributions can then be used to assess the susceptibility or resistance of each family.

Table III Empirical powers for the data set under consideration

Percentage of families			Relative odds ratio	Power	
Pre-disposed to disease	Randomly affected by disease	Resistant to disease		For 52 families	For 206 families
25	50	25	3,0	34%	80%
50	0	50	1,5	10%	*
50	0	50	3,0	68%	*

* not calculated

4. THE POWER OF THE PROCEDURE

The power of the proposed procedure depends on many factors, such as the number of families in the sample, the family sizes, the age and sex composition of the families, the probability of contracting the disease and, most importantly, the number of families either predisposed to or resistant to

the disease, as well as the degree of familial predisposition or resistance. The power of the procedure therefore represents a complex concept about which, in general, not much can be stated. It is possible, however, to calculate the empirical powers for a number of selected combinations of the most important factors mentioned above for a specific data set. Certain subsets of families within the available data set can therefore be defined as being either predisposed or resistant to the disease. It is also possible to define the degree of predisposition or resistance to the disease, for example, three times more than random, i.e. a relative odds ratio of 3, etc. This approach represents an over-simplification of the practical situation where the degree of predisposition or resistance can vary from one family to the next. As stated in the Introduction, this variable may even follow a continuous distribution.

The procedure for calculating the empirical power is as follows:

1. Decide on the proportion of families to be regarded either as predisposed or resistant to the disease. Then decide on the risk ratio between predisposed and randomly affected families, as well as between resistant and randomly affected families.
2. With the percentage and relative odds ratios determined in step 1, calculate the probabilities to contract the disease for the random, predisposed and resistant groups of families for each age/sex category, in such a way that the overall frequency of disease manifestation within the data set remains unchanged.
3. The simulation is as before.

5. RESULTS

All calculations were done with SAS statements and procedures. The number of $\hat{Q}_0$ -values simulated for this data set is $N = 300$. The empirical percentiles calculated are as follows:

$$P_{95} = 0,025; \quad P_{99} = 0,027; \quad \text{Median} = 0,020.$$

The Q-value for this data set is calculated as $\hat{Q} = 0.0393$. H_0 is defined as follows: there is no familial predisposition or resistance to the disease. As can be seen from the empirical percentiles calculated, this hypothesis can be rejected at the 1%-level of significance. Hence it is reasonable to conclude that a familial predisposition and/or a resistance to the disease is present in this particular data set of families.

The empirical power of the above-mentioned test procedure has also been investigated for the specific composition of the 206 available families. The results of this investigation are presented in table III. From these it is evident that, as expected, the power increases,

- if the relative odds ratio between predisposed and resistant families increases,
- if the proportion predisposed and/or resistant families increases, and
- if the number of families within the sample is increased.

The latter factor, in particular, can be seen to have a dramatic effect on the power.

The empirical powers for the data set under consideration are set out in table III. From this tentative investigation of the power, it appears that the proposed procedure is suitable for identifying predisposition and/or resistance within a sample of families, if at all present.

6. DISCUSSION

As stated in the Introduction, the objective of the present theoretical investigation is to develop an objective statistical method for the quantification of a positive or negative family history, i.e. either a significant familial predisposition or resistance to a certain disease condition. The method realistically accommodates and weighs the most important parameters which underlie and determine a family predisposition (such as specified in the context of the theoretical data, see above), and expresses this in the form of a single characteristic risk index for each specific family.

These risk indices of families then can be evaluated and quantified relative to the overall "average" risk index derived from the sample of families, and it permits the classification or ranking of families in terms of their predisposition or resistance to the disease. Provision is also made for testing of the hypothesis that the disease manifests itself at random.

The statistical procedure as outlined is computer intensive in the sense that much use is made of c.p.u.-time. If the number of families within the sample is not much larger than 200, however, the procedure is quite feasible. It is also evident that the power in the case of about 200 families is of an acceptable order of magnitude. For small odds ratios (less than 3), larger samples of families may be required to ensure an acceptable power. These observations regarding the power are based on the limited information derived from the single sample of families used as an example in this paper. Additional studies would be required before more specific statements can be made regarding the power of the procedure.

As explained above, the expected value $E(\hat{I}_k)$ of the proposed family index $\hat{I}_k$ is zero if the disease condition occurs randomly in a particular family. Indeed, if we assume a disease to be highly sex and age dependent in its manifestation, then families with an "average" sex and age distribution and an "average" number of affected individuals will have their $\hat{I}_k$ -value close to zero. In families with many affected individuals $\hat{I}_k$ will tend to be highly positive, and even more highly positive in families with many affected young females (assuming late onset and preponderance in males, to be the typical pattern of manifestation for the specific disease), and conversely.

A situation where all families in a sample are either homogeneously predisposed or homogeneously resistant to a disease will not be amenable to the present method of analysis. Such a situation can be visualized as a random variation of uniform "familial predisposition" within a sample of families, all of which either are predisposed or resistant to the disease condition.

Although the proposed technique includes a test of a null hypothesis, it must be emphasized that the test is empirical. The technique, therefore, is rather explorative in its application and does not provide a confirmatory

statistical test.

In the derivation of our procedure the probability distributions for age and sex were calculated from the relative frequencies of these variables within the data set under investigation (see tables I and II). These probability distributions can, however, also be derived from an external source, such as the incidence of the disease by sex and age group within the entire population.

It must be emphasized that the familial predisposition or resistance to a disease condition may be genetic and/or environmental in origin. The statistical method presented makes no distinction between these etiological contributors, but quantifies them jointly, i.e. the effect of the common "familial" genotype and of the "familial" environment, as shared to a larger or lesser degree by all family members. Should it be necessary to specifically distinguish between, and separately estimate, the genetic and environmental effects, other approaches and methods will have to be employed. However, our statistical technique is suitable for comparing a certain test set with a control set of environmental factors regardless of a certain environment.

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